Huawei Access Node WHI-ACC-81 Hardware/Environmental Fault Resolution Procedure

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4446

Description

This procedure outlines the steps to resolve hardware/environmental faults on Huawei Access Node WHI-ACC-81 at the Whitefield East POP, specifically for the role of Access. The failure domain of interest is Hardware/Environmental Fault, characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. This situation is likely to occur due to the inherent complexity of the ASIC-based architecture, where memory parity errors can lead to catastrophic failures of the line card and subsequent loss of power supply redundancy. In Huawei's CLI dialect, the role of Access is denoted by the [Access] command, and the [display fault-manage] command is used to identify potential issues. The resolution procedure involves isolating the faulty components, replacing them if necessary, and verifying the system's functionality post-resolution. It is essential to adhere strictly to the CLI dialect and follow the specified steps to avoid any further damage to the system.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Whitefield East POP to assist with the resolution.

- Check the status of the redundant link to ensure it is functioning properly.

- Obtain the necessary replacement parts for the faulty components.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-manage, Expected Output: Fault management information, Description: This command is used to display fault management information, including alarm messages, error messages, and fault logs. It helps to identify potential issues and isolate the source of the problem.

Command: display error-indication, Expected Output: Error indication information, Description: This command is used to display error indication information, including error messages, alarm messages, and error logs. It helps to identify potential issues and isolate the source of the problem.
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Command: display power-supply, Expected Output: Power supply information, Description: This command is used to display power supply information, including power supply status, power consumption, and power supply redundancy. It helps to identify potential issues with the power supply system.
```

Resolution Steps

Step 1: Step: Isolate the faulty component, Command: display fault-manage, Description: Use the display fault-manage command to identify the faulty component and isolate it from the rest of the system.

Step 2: Step: Replace the faulty component, Command: replace [component name], Description: Once the faulty component has been isolated, replace it with a new one. Use the replace command to specify the component name and the replacement part number.

Step 3: Step: Verify the system's functionality, Command: display system-information, Description: Use the display system-information command to verify that the system is functioning properly post-resolution.

Post-Check Verification

```
Command: display system-information, Expected Output: System information, Description: This command is used to display system information, including system status, system configuration, and system logs. It helps to verify that the system is functioning properly post-resolution.

Command: display fault-manage, Expected Output: Fault management information, Description: This command is used to display fault management information, including alarm messages, error messages, and fault logs. It helps to verify that the system is functioning properly post-resolution.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: undo [configuration command], Description: If the resolution fails, restore the original configuration using the undo command. This helps to safely rollback the changes made during the resolution procedure.

- Step: Reboot the system, Command: reboot, Description: If the resolution fails, reboot the system to return it to its original state. This helps to safely rollback the changes made during the resolution procedure.